Fiscal Year:	FY 2020	**Task Last Updated:**	FY 09/02/2020
PI Name:	Main, Bob M.S.		
Project Title:	Development of Advanced Vision Testing Software for a Goggle-Based Visual Field Device Prototype		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bob@bobmain.net	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	801-979-2150
Organization Name:	Web Vision		
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PI Address 2:			
PI Web Page:			
City:	South Jordan	**State:**	UT
Zip Code:	84009	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Focused Investigations
Start Date:	08/01/2018	**End Date:**	05/31/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2020 (originally 7/31/2019) per E. Urquieta/TRISH (Ed., 9/18/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Buscemi, Philip O.D. (Buscemi Group, LLC)		
Grant/Contract No.:	NNX16AO69A-FIP0009		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Focused Investigation Project Web Vision will design and develop (working with outside contractors) the software for four vision tests that will be used with the Goggle-Based Visual Fields (GBVF) device currently begin developed based on NASA specifications and with TRISH (Translational Research Institute for Space Health) funding. This project will take 1 year to complete. We have been and will continue to work closely with NASA Spaceflight-Associated Neuro-ocular Syndrome (SANS) Team to insure the vision tests being developed meet/exceed their needs, requirements, and specifications. A final version of the four enhanced vision tests for the GBVF device will be delivered (and demonstrated) to NASA one year after the project begins.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Device will allow NASA astronauts to easily test for blind spots in the visual field in a small footprint providing valuable astronaut visual health information to flight surgeons monitoring early stages of SANS.
Task Progress:	Project Aim/Objective: Web Vision is currently developing a prototype Goggle-Based Visual Fields (GBVF) testing device with TRISH funding. This project is to develop advanced vision testing capabilities (software) for that Goggle-Based Visual Fields device. Original project statement of work (SOW) goals: 1) kinetic papillary detection, 2) panoramic visual field, 3) contrast sensitivity testing, 4) advanced version of the Ishihara color test. What we delivered: Based on NASA request, we developed Advanced Software for the Goggle Based Visual field device (GBVF). The three elements of the software are: 1) gaze tracking support, 2) improved blind spot detection, and 3) faster/adaptive threshold strategy. These software improvements are already included in the GBVF device (delivery late July 2020). SOW changed August 2019 at NASA request to: 1) gaze tracking support, 2) improved blind spot detection, and 3) faster/adaptive threshold strategy. No additional improvements for this project are planned. We have submitted software improvements in the GBVF device project (PHASE 2).
Bibliography Type:	Description: (Last Updated:)